

BOOM: Carbohydrates and Carbohydrate Metabolism Overview I (2 of 2)

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Learning Objectives (3)

After completing this brick, you will be able to:

- 1 Describe the importance of the structural and dynamic nature of glycogen stores.
- 2 Explain the biochemical basis for the cause and symptoms of lactose intolerance and galactosemia.
- 3 Outline the substrates, products, and enzymes of the pentose phosphate pathway (PPP).



CASE CONNECTION

You are asked to serve on a state panel to review a proposal from a group of citizens who are against state-mandated neonatal screening and vaccines.

“Parents are the ones who should make these decisions for their children which could also save taxpayer dollars,” one person stated. “For example, galactokinase deficiency is on the required screen and I read online this is referred to as a mild condition. If this is the case, why are we forced to have our children tested for it?”

How will you describe the importance of neonatal screening for galactokinase deficiency? Consider your response as a medical professional, and we'll revisit the scenario at the end of the brick.

Describe the importance of the structural and dynamic nature of glycogen stores (LO#1)

Glycogen is a highly branched glucose polymer contained within the liver and skeletal muscle. Glycogen serves as a critical depot for excess glucose. While the liver will liberate and export individual glucose molecules derived from glycogen to support the metabolism in other tissues, the glycogen stored within skeletal muscle will be utilized by skeletal muscle alone.

pancreas and basal side of enterocytes in the intestine.
GLUT2 is a reversible glucose transporter expressed on liver,

Glycogen has a linear core of glucose linked via α -1,4 glycosidic bonds with branching occurring every 8-12 glucose units. These branch points uniquely contain α -1,6 glycosidic links with subsequent glucose within the branches connected again by α -1,4 glycosidic bonds (**Figure 1**).

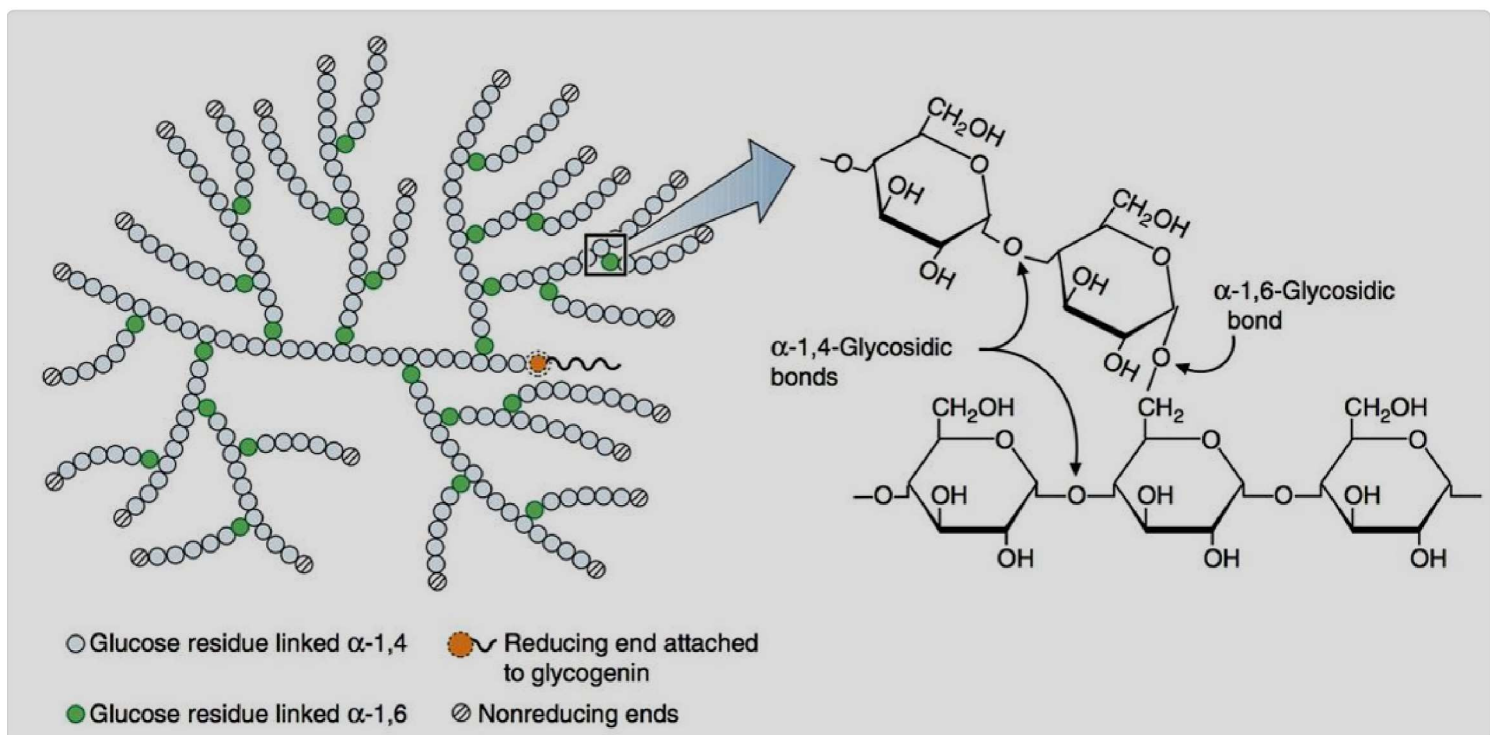


Figure 1. Glycogen structure with branch points.

The biological benefit of such a highly branched glucose storage structure is two-fold:

1. One massive structure of many glucosyl units versus individual glucose molecules of the same number, greatly minimizes osmotic effects of intracellular glucose. Having the same amount of glucose as individual

monosaccharides instead of polymerized glycogen would cause a massive influx of water in an attempt to balance extracellular and intracellular osmolarity. This would cause the tissue to swell, distort, and compromise function.

2. The highly branched structure permits rapid glucose release from glycogen stores as it provides multiple terminal ends from which glucose can be liberated (e.g. in exercising muscles or liver during times of fasting degradative enzymes can act on each non-reducing end simultaneously). If it were linear, the structure would only provide one site from which to "pluck" a single glucose. The additional non-reducing ends of glycogen provide multiple sites from which to release glucose increasing the concentration far more rapidly.



CLINICAL CORRELATION

For the highly branched glycogen to be an efficient glucose storage mechanism, the glucose must be incorporated into the branches during time of plenty and extracted when glucose is needed. For this purpose, there are specialized enzymes within skeletal muscle and liver that form (i.e. branching enzymes) and destroy (i.e. debranching enzymes) glycogen. Deficiency of these enzymes falls under a class of disorders known as glycogen storage diseases (GSD). Andersen disease (i.e. type IV GSD) is caused by a branching enzyme deficiency. The core structure of the glycogen is formed, but the branches fail to form and cause major liver dysfunction. This is largely due to an immune response as the abnormal glycogen structure is perceived as a foreign body creating chronic hepatic inflammation. Due to glucose imbalance and advanced liver cirrhosis, this condition is usually fatal within 2 years. We will consider all the GSDs at length later in the course.

α -1,4 and α -1,6, respectively

Explain the biochemical basis for the cause and symptoms of lactose intolerance and galactosemia (LO#2)

Low expression or deficiencies in enzymes that metabolize sugars can have significant consequences. We will explore two initial examples in which individuals fail to properly digest a disaccharide and another in which a simple monosaccharide that comprises the disaccharide is unable to be metabolized. Each have distinct presentations, deficiencies, and treatment strategies.

Lactose Intolerance

When lactose is ingested, it is normally cleaved into monosaccharides (i.e. glucose and galactose) by the lactase enzyme with subsequent absorption into the intestinal wall via SGLT. However, when lactase is deficient, lactose

remains available in the gut lumen for bacterial metabolism (**Figure 2**). Once bacteria feed on lactose, they release gases such as CO_2 and CH_4 , lactic acid, and other metabolic byproducts. The gas results in bloating and flatulence while the lactic acid is osmotically active. The increased production of lactic acid causes an influx of water into the gut lumen. This causes subsequent distention of the intestinal wall which produces powerful peristaltic contraction resulting in watery diarrhea. This paradigm is summarized in **Figure 2**.

In contrast to what is commonly seen as an inconvenience in adults (i.e. adult lactase deficiency, aka adult hypolactasia), congenital lactase deficiency is a severe autosomal-recessive inherited disease in which lactase activity is significantly reduced, or totally absent. The disorder presents as soon as the newborn is fed breast milk or lactose-containing formula, resulting in watery diarrhea, weight loss, and dehydration. Treatment consists of removal of lactose from the diet, which allows for normal growth and development to occur.

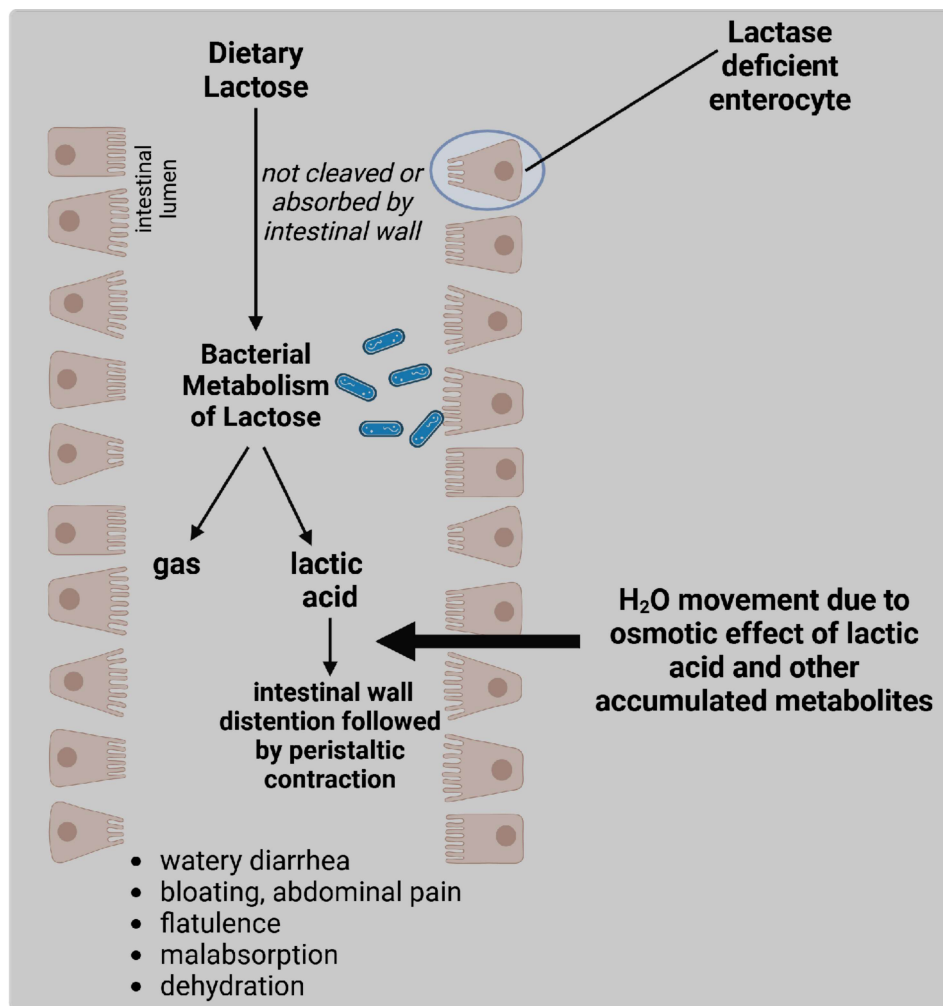


Figure 2. Lactose intolerance.

CLINICAL CORRELATION

The hydrogen breath test is a non-invasive diagnostic tool used to identify various gastrointestinal conditions, such as lactose intolerance, SIBO (small intestine bacterial overgrowth), and carbohydrate malabsorption as can occur with fructose. A patient is provided a specific carbohydrate to consume, and breath samples are analyzed over time after ingestion for the presence of hydrogen, methane, or hydrogen sulfide. Elevated gas content in the sample

indicates that undigested or unabsorbed sugars are being fermented by bacteria in your gut.



CLINICAL REASONING

The majority of the world's population is lactose intolerant to some degree. In most individuals, lactase expression increases during gestation. Lactase levels remain high for ~1 month after birth to ensure proper digestion of milk and then lactase enzyme expression begins to decline. For an average person, this decline is fairly steady until about an age of 5 when levels become <10% of those present in infants. This is the typical residual activity an individual will have throughout adulthood. Genetic ancestry greatly influences lactase expression levels in adults. For example, individuals of Native Indian genetic descent in Oklahoma have a very high incidence of lactase deficiency in adulthood (>95%). In contrast, Danish and Dutch populations exhibit less than 3% on average of individuals with lactose intolerance.

Data from Biller HA, Grand RS. Lactose intolerance. *Annu Rev Med.* 1990;41:141–148.

Galactose metabolism

If not properly metabolized, ingestion of galactose can have serious health consequences. Humans primarily obtain galactose in the form of lactose, which is present naturally in milk or dairy products, including breast milk and most infant formulas. The liver predominantly metabolizes galactose while to a lesser extent the intestines and kidney also metabolize galactose.

The critical enzymes in galactose metabolism are:

- **Galactokinase:** Highly efficient enzyme that converts galactose to galactose 1-phosphate. Galactokinase has a fairly low K_m (i.e. high affinity) but a high V_{max} . This step consumes ATP to provide the phosphorylated galactose. Importantly, galactose-1-phosphate accumulation as the main toxic mediator in Type I.
- **Galactose 1-phosphate uridyl transferase (GALT):** Converts galactose 1-phosphate to glucose 1-phosphate by switching the attached groups of galactose 1-phosphate and UDP-glucose. Note: some sources will refer to GALT as uridylyl, but they refer to the same enzyme.
- **Aldose reductase:** This enzyme is usually not physiologically active in galactose metabolism. However, if galactose is in excess, this enzyme will aberrantly convert galactose to galactitol consuming NADPH in the process. Aldose reductase has a very high K_m (i.e. low affinity) and is only active when substrates excessively accumulate within a cell.

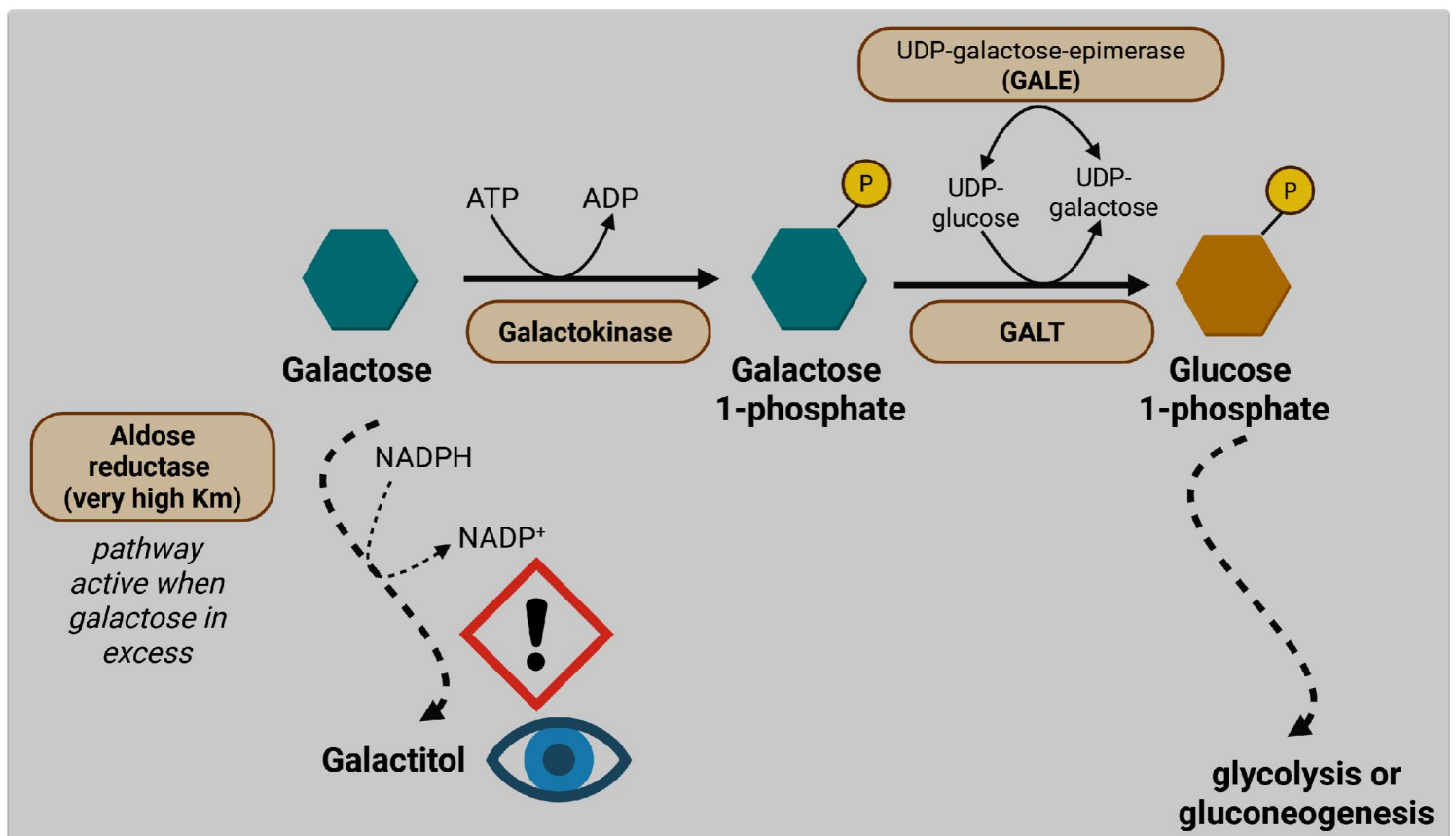


Figure 3. Galactose metabolism. Enzyme deficiencies in this pathway cause increased circulating galactose (galactosemia). In these conditions, excess galactose spills over into other tissues such as the lens of the eye to promote galactitol production via aldose reductase.

Critical Note: There is no natural enzyme that breaks down galactitol, nor is there a transporter that allows galactitol to exit the cell. Therefore, cellular accumulation causes osmotic stress. When galactitol is produced in the pathway, NADPH is also consumed (**Figure 3**). This provides less NADPH to ensure proper levels of reduced glutathione (i.e. GSH) are present to neutralize harmful oxidative species. These adverse conditions cause both osmotic and oxidative stress that denatures protein and result in cellular damage in the liver, eyes, and brain.

There are two major galactose metabolism disorders: GALT deficiency, also known as type I galactosemia (i.e. classic galactosemia) and galactokinase deficiency (i.e. type II galactosemia).

Classical Galactosemia (Type I)

The more severe of the two galactose metabolism disorders is classical galactosemia in which there is a deficiency of the **galactose 1-phosphate uridyl transferase (GALT)** enzyme (**Figure 3**). GALT deficiency causes galactose 1-phosphate and galactose to accumulate. As the first step is intact, it depletes the cell of ATP and causes phosphate trapping. This loss of ATP impairs hepatic gluconeogenesis that requires ATP. High levels of galactose 1-phosphate also inhibits glycogen phosphorylase, the enzyme required for the first step of glycogen degradation. Inhibition of metabolic cascades that ensure proper blood glucose levels during times of fasting leads to:

- hypoglycemia

- GI distress (e.g. vomiting, diarrhea)
- weakness
- hepatomegaly
- jaundice

Patients with classical galactosemia present in infancy (usually in the first few days of life **after** breastfeeding) with failure to thrive (i.e. failure to grow and gain weight at the expected rate). If not treated with dietary intervention, this enzyme deficiency leads to long term complications such as:

- developmental delays
- intellectual disabilities
- liver cirrhosis



CLINICAL CORRELATION

A similar ATP depletion and phosphate trapping occurs in hereditary fructose intolerance which is a deficiency of the enzyme aldolase B. Much like galactokinase, fructokinase is an extremely efficient enzyme that phosphorylates fructose by rapidly consuming ATP. We will consider this disorder in fructose metabolism and how it relates to galactosemia in the next sessions.

Perhaps, most notably, patients develop **cataracts** at a very young age. Deficiencies in classic galactosemia cause high levels of circulating and excreted galactose when consumed. Cataracts occur because the excess galactose is converted to galactitol in the lens through the aldose reductase

enzyme (**Figure 3**). The build-up of galactitol produces osmotic and oxidative stress in the lens which denatures and aggregates proteins leading to opaqueness. A similar process occurs with diabetes wherein excess glucose is converted to sorbitol. Fortunately, most cases of classical galactosemia are now picked up by newborn screening, and proper dietary strategies/interventions can be implemented early to prevent complications.

Galactokinase Deficiency (Type II)

Compared with classical galactosemia, galactokinase deficiency is a mild disease. Patients will have elevated levels of galactose in the blood and urine. Like the classic form, excess galactose promotes galactitol formation in various tissues of the body through the action of aldose reductase. Therefore, the main clinical manifestation of galactokinase deficiency is cataracts, which usually presents in childhood or young adulthood. There could be signs of visual impairments early in infancy such as a failure to track objects or a visual recognition of caregivers by a smile, laugh, etc. However, the severe symptoms that occur with classical galactosemia such as liver dysfunction or mental disability are **lacking**.

Elimination of galactose-containing foods, primarily dairy products, but other sources include organ meat (e.g. pancreas, liver) and chickpeas (garbanzo beans), is the mainstay of treatment for galactose metabolism disorders. Individuals with galactosemia should follow a strict diet such as drinking lactose-free soy-based infant formulas as infants. A lactose and galactose free diet prevents all complications related to the enzyme deficiency.

Critical Note: Type III is a far rarer form of galactosemia. This form is caused by a lack of UDP-galactose-epimerase (GALE) enzyme that reversibly isomerizes the UDP-galactose formed during galactose metabolism to UDP-

glucose that can be used again in the pathway (**Figure 3**). The severity is similar to classic galactosemia (type I) as the phosphorylated galactose can still be formed. However, select variants have a milder presentation.



CLINICAL REASONING

Human breast milk generally lacks fructose and sucrose but contains lactose. This is why galactosemias typically present much earlier after birth than disorders in fructose metabolism. Interestingly, even someone who is lactose intolerant and restricts consumption of lactose, is still capable of producing lactose within mammary tissue. Mammary glands express lactose synthase which enzymatically converts UDP-galactose and glucose into the lactose disaccharide and free UDP. Remember from the galactosemia scheme (**Figure 3, GALE reaction**), UDP-galactose is interconverted to UDP-glucose and vice versa by GALE. Therefore, individuals that have galactosemia will also produce lactose in their milk despite limiting galactose in their diet. This demonstrates that a lactating mother does not need to consume lactose or galactose for lactose to be produced in the milk.



CLINICAL CORRELATION

Excess galactose inside neutrophils impairs their bactericidal activity, so newborns are at risk for *Escherichia coli* sepsis. This connection to immunology and microbiology is an interdisciplinary point often tested on board exams.

The following question must be completed before continuing

ACTIVE LEARNING QUESTION

Match the descriptor with the correct enzyme or enzyme deficiency.

produces a sugar alcohol known as galactitol but only when galactose levels are high

required for producing galactose and glucose from a dietary disaccharide

deficient in type II galactosemia which is considered milder

is associated with hypoglycemic states due to inhibited glycogen degradation and gluconeogenesis

isomerizes UDP-galactose into UDP-glucose

lactase	
aldose reductase	
galactokinase	
galactose-1-phosphate uridyl transferase (GALT)	
UDP-galactose-epimerase (GALE)	

Explanation (requires correct answer)

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